

98. If $\frac{x}{y} = \frac{6}{5}$, then the value of $\left(\frac{6}{7} - \frac{5x-y}{5x+y}\right)$ is equal to
- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{3}{7}$ (d) $\frac{4}{7}$
99. If $\frac{x}{y} = \frac{1}{3}$, then $\frac{x^2 + y^2}{x^2 - y^2} = ?$ (R.R.B., 2006)
- (a) $-\frac{10}{9}$ (b) $\frac{5}{4}$
 (c) $-\frac{5}{4}$ (d) $-\frac{5}{3}$
100. If $\frac{x}{2y} = \frac{6}{7}$, the value of $\frac{x-y}{x+y} + \frac{14}{19}$ is
- (a) $\frac{13}{19}$ (b) $\frac{15}{19}$
 (c) 1 (d) $1\frac{1}{19}$
101. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, then $\frac{c^2 - a^2}{c^2 + a^2}$ is (M.B.A., 2007)
- (a) $\frac{1}{7}$ (b) $\frac{7}{25}$
 (c) $\frac{3}{4}$ (d) None of these
102. If $x = 1 - q$ and $y = 2q + 1$, then for what value of q , x is equal to y ?
- (a) -1 (b) 0
 (c) $\frac{1}{2}$ (d) 2
103. Find x if $\frac{x}{5} - \frac{x}{6} = 4$.
- (a) -120 (b) -100
 (c) 100 (d) 120
104. If $4x + 5y = 83$ and $\frac{3x}{2y} = \frac{21}{22}$, then $y - x = ?$
- (a) 3 (b) 4
 (c) 7 (d) 11
105. If $a = \frac{x}{x+y}$ and $b = \frac{y}{x-y}$, then $\frac{ab}{a+b}$ is equal to
- (a) $\frac{xy}{x^2 + y^2}$ (b) $\frac{x^2 + y^2}{xy}$
 (c) $\frac{x}{x+y}$ (d) $\left(\frac{y}{x+y}\right)^2$
106. If $\frac{a}{b} = \frac{1}{3}$, $\frac{b}{c} = 2$, $\frac{c}{d} = \frac{1}{2}$, $\frac{d}{e} = 3$ and $\frac{e}{f} = \frac{1}{4}$, then what is the value of $\frac{abc}{def}$? (C.A.T., 2006)
- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
 (c) $\frac{3}{8}$ (d) $\frac{27}{4}$
 (e) $\frac{27}{8}$
107. If $\frac{m}{n} = \frac{4}{3}$ and $\frac{r}{t} = \frac{9}{14}$, the value of $\frac{3mr - nt}{4nt - 7mr}$ is (M.B.A. 2011)
- (a) $-5\frac{1}{2}$ (b) $-\frac{11}{14}$
 (c) $-1\frac{1}{4}$ (d) $\frac{11}{14}$
108. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$, then $c + \frac{1}{a}$ is equal to (S.S.C., 2007)
- (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) 2
109. If $y + \frac{1}{z} = 1$ and $x + \frac{1}{y} = 1$, then what is the value of xyz ?
- (a) -1 (b) 0
 (c) $\frac{1}{2}$ (d) 1
110. If $\frac{x}{(2x+y+z)} = \frac{y}{(x+2y+z)} = \frac{z}{(x+y+2z)} = a$, then find a if $x + y + z \neq 0$. (M.A.T., 2005)
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
111. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = K$, then the value of K is
- (a) $\pm \frac{1}{2}$ (b) $\frac{1}{2}$ or -1
 (c) -1 (d) $\frac{1}{2}$
112. If $a - 8 = b$, then determine the value of $|a - b| - |b - a|$.
- (a) 0 (b) 2
 (c) 4 (d) 16